



ApexaiQ, IT Asset Management & Cybersecurity Research

Executive summary

- **ApexaiQ is a continuous asset-assurance platform.** It brings together asset discovery, inventory quality, vulnerabilities, obsolescence, maintenance, compliance context, prioritization, and remediation in one view.
- **The problem it addresses is not simply “too many devices.”** It is the gap between what an organization believes it owns and what is actually connected, exposed, unsupported, misconfigured, or business-critical.
- **Its score is a management signal, not a universal security rating.** ApexaiQ describes the ApexaiQ score as a 60–160 scale inspired by IQ. The score combines the IT environment, asset hygiene, and IT gaps; the exact weighting is not publicly documented.
- **Agentless is a deliberate design choice.** It avoids installing and maintaining software on every endpoint, which is useful for unmanaged, legacy, IoT, OT, network, cloud, and third-party assets. The trade-off is that agentless discovery depends on network visibility, integrations, credentials where required, and the limits of the protocols or data sources available.
- **The closest comparison is a blend of ITAM, CAASM, attack-surface management, vulnerability context, and remediation workflow.** Axonius, Armis, runZero, and JupiterOne overlap with parts of this value proposition, but each emphasizes a different center of gravity.

1. What ApexaiQ does

ApexaiQ positions itself as a SaaS-based, agentless, continuous asset-assurance platform. In practical terms, it aims to answer six questions:

1. **What assets exist?** Hardware, software, firmware, cloud resources, users, access controls, and connected devices.
2. **Which records refer to the same asset?** Data is deduplicated and normalized so teams are not working from conflicting inventories.
3. **What is the condition of each asset?** The platform enriches assets with vulnerability, maintenance, warranty, obsolescence, and compliance information.
4. **Which assets matter most to the business?** Criticality and business context help separate a serious exposure on a crown-jewel system from a low-impact issue on a disposable test device.
5. **What needs attention first?** The product prioritizes remediation instead of presenting a flat list of findings.
6. **Can the organization prove that risk was reduced?** Reporting, benchmarks, audit evidence, and closed-loop remediation help connect discovery to action and verification.

The underlying industry problem is **fragmented and unreliable technology visibility**. Asset data often lives in a CMDB, endpoint management tool, vulnerability scanner, directory, cloud console, ticketing system, spreadsheet, or a managed service provider's platform. These systems rarely agree perfectly. The result is an environment with:

- unknown or shadow assets;
- duplicate or stale records;
- vulnerable assets that cannot be mapped to an owner;
- devices that are out of warranty or beyond vendor support;
- software and firmware that are difficult to patch;
- audit preparation that depends on manual evidence collection; and
- security teams forced to prioritize without reliable business context.

ApexaiQ's value proposition is to turn that fragmented picture into a continuously updated, risk-oriented view. That is broader than traditional inventory management and narrower than a full SIEM, EDR, or network security control. It

helps teams decide what to fix, but it does not replace every control that detects, blocks, authenticates, or responds to an attack.

2. The ApexaiQ score

ApexaiQ describes its score as a numerical view of overall infrastructure health and vulnerability. Public product material describes a range from **60 to 160**, inspired by human IQ: lower scores represent weaker conditions and higher scores represent stronger, more secure environments.

The score is described as combining three areas:

Input	What it represents	Why it matters
IT environment	The shape and condition of the technology estate	A small, stable environment has a different risk profile from a large, distributed estate with cloud, IoT, remote, and legacy assets
Asset hygiene	Obsolescence, maintenance, and vulnerabilities	These are recurring indicators of technical debt and exploitable exposure
IT gaps	Missing controls, non-compliance, or unmanaged conditions	Gaps show where the intended security or operating standard is not being met

The most useful way to use the score is as a **trend and prioritization signal**:

- establish a baseline;
- identify which categories are pulling the score down;
- remediate the highest-impact issues;
- measure whether the score and the underlying evidence improve; and
- explain the change to technical and executive audiences.

The score should not be treated as a replacement for CVSS, the CISA KEV catalog, a formal risk assessment, an ISO/IEC 27001 audit, or an incident-response capability. A single number can make direction and progress easier to communicate, but the supporting asset and control evidence is what makes the number actionable. ApexaiQ does not publicly disclose the full scoring formula, weights, calibration data, or independent validation, so comparisons should be

made within the same organization and over time rather than against an assumed industry-wide standard.

3. IT asset management: what it is and why companies need it

IT asset management (ITAM) is the disciplined management of technology assets across their lifecycle: plan, acquire, deploy, configure, operate, maintain, secure, renew, transfer, retire, and dispose.

An IT asset may be physical, virtual, cloud-based, software-based, or a service. Examples include laptops, servers, routers, firewalls, switches, phones, virtual machines, containers, SaaS subscriptions, applications, databases, certificates, firmware, identities, and data stores.

A mature ITAM practice answers more than “where is the laptop?” It also answers:

- Who owns or is responsible for it?
- What software, version, firmware, and configuration does it have?
- Is it authorized?
- Is it exposed to the internet or connected to a sensitive segment?
- Is it supported and under warranty?
- Which vulnerabilities affect it?
- What business process or data depends on it?
- What controls should apply?
- What happens when it is retired?

NIST’s IT asset management guidance describes the value of connecting physical and virtual assets so an organization can understand what it has, where it is, how it is used, and which assets are vulnerable. That connection improves both security analysis and asset utilization.

Companies need asset-management software because spreadsheets and periodic audits do not keep pace with modern environments. Software can:

- reconcile records from multiple tools;

- identify duplicates, gaps, and stale data;
- discover unauthorized or previously unknown assets;
- link vulnerabilities to actual devices and owners;
- track lifecycle, warranty, support, and maintenance status;
- automate notifications, tickets, and remediation jobs;
- support audit evidence and regulatory reporting; and
- give finance, procurement, IT operations, security, and leadership a shared source of truth.

ITAM is therefore not only an IT operations discipline. It is a foundation for cybersecurity, resilience, procurement, audit readiness, business continuity, and cost control.

4. Why ApexaiQ is agentless

An **agent** is software installed on an endpoint or workload to collect telemetry or enforce controls. An **agentless** platform gathers information through network discovery, APIs, directories, cloud and SaaS integrations, management systems, scanners, logs, and standard protocols rather than requiring an ApexaiQ agent on every device.

ApexaiQ's agentless model is useful for several reasons:

- **Coverage of unmanaged assets.** An organization cannot install an agent on every printer, switch, firewall, medical device, industrial controller, guest device, contractor device, or shadow asset.
- **Lower deployment friction.** Teams avoid packaging, testing, updating, and troubleshooting another endpoint component.
- **Less endpoint overhead.** There is no additional resident process competing for CPU, memory, storage, or administrative attention on every machine.
- **Faster acquisition and integration.** Existing sources can be connected instead of waiting for a complete endpoint-agent rollout.
- **Better fit for sensitive or fragile systems.** Some legacy, OT, embedded, or regulated devices cannot safely run third-party software.

- **Reduced blind spots during mergers and due diligence.** Network and integration-based discovery can start before a target environment is fully standardized.

Agentless does not mean “no collection infrastructure” and it does not mean “no access requirements.” The platform still needs a path to the data: network reachability, APIs, connectors, read-only credentials, directory access, cloud permissions, or other telemetry. It can also provide less deep host-level detail than a specialized endpoint agent. The design choice is a trade-off: broader and faster coverage of the estate, with depth depending on the source and protocol.

5. Competitive landscape

These competitors overlap with ApexaiQ, but they are not identical products. The comparison below is about market emphasis, not a claim that one platform is universally better.

Platform	Center of gravity	How it differs from ApexaiQ	Best fit when
Axonius	Integration-led cyber asset management and asset intelligence	Typically starts by ingesting data from many existing tools, reconciling it into an authoritative asset model, and automating actions. ApexaiQ puts more visible emphasis on continuous asset assurance, hygiene, lifecycle context, and an executive-facing score.	The organization already has many security and IT tools and needs one reconciled asset layer plus automated enforcement workflows
Armis	Agentless asset intelligence and real-time security for IT, IoT, OT, medical, and unmanaged devices	Armis is especially strong in device behavior, threat detection, and broad cyber-physical visibility. ApexaiQ is more explicitly oriented toward hygiene, lifecycle, compliance, prioritization, and asset-assurance reporting.	The environment includes large numbers of unmanaged or cyber-physical assets and real-time threat context is central

Platform	Center of gravity	How it differs from ApexaiQ	Best fit when
runZero	Active network discovery and credentialless or unauthenticated asset discovery	runZero is known for fast network discovery of managed and unmanaged assets, with a strong discovery and exposure-management focus. ApexaiQ extends further into maintenance, obsolescence, compliance, scorecards, and remediation context.	The immediate problem is “what is really on the network?” and fast discovery is more important than a broad lifecycle program
JupiterOne	Graph-based CAASM, exposure management, queries, and continuous controls monitoring	JupiterOne emphasizes relationships among cloud, code, identities, applications, vulnerabilities, and infrastructure, with a query-driven operating model. ApexaiQ emphasizes normalized asset assurance and a simpler score-and-action narrative.	Security teams want to ask complex questions across a connected asset graph and operationalize continuous control monitoring

Case-study snapshots

The following outcomes come from vendor-published case studies and should be treated as directional evidence rather than independently audited benchmarks.

- **ApexaiQ and a regional bank:** A published case study describes a four-week implementation using Active Directory, Coda, Defense Storm, and GFI. The bank used the platform for a single CMDB view, cleaner audit and cybersecurity-insurance data, vulnerability prioritization, automated tagging, and ticketing integration. The stated result was less audit-preparation effort, improved auditor confidence, and better operational efficiency.
- **Axonius and a global specialty retailer:** The retailer reported discovering unauthorized or suspicious backdoor software on 25 computers and using

automated asset intelligence to understand the scope and impact of urgent issues.

- **Armis and a medical-device manufacturer:** The case study describes incomplete visibility across IT, IoT, and OT assets, complicated by acquisitions. The organization used agentless discovery to build a more complete inventory, improve vulnerability management, and align asset understanding with NIST CSF practices.
- **runZero and Capgemini:** A luxury manufacturer and retailer needed accurate discovery without credentials. runZero reportedly found that the existing inventory had been overestimated by 33%; it also reported scanning a retail store in under two minutes, a small data center in under six minutes, and a large data center in about 30 minutes.
- **JupiterOne and an American financial-services company:** The organization had siloed scanners and manual spreadsheet-based correlation. JupiterOne reported reducing vulnerability-to-asset correlation work by 20x, moving discovery from as much as two days to roughly five minutes, and giving teams a centralized, self-service asset and vulnerability view.

Practical competitive conclusion

ApexaiQ is most differentiated when the buyer wants a **continuous, agentless asset-assurance program** that combines inventory quality with lifecycle risk, compliance context, a clear score, and remediation tracking. It may be less differentiated when the primary requirement is deep endpoint telemetry, real-time behavioral detection, or highly specialized network discovery. In those situations, ApexaiQ may complement rather than replace an EDR, NDR, vulnerability scanner, CMDB, or SIEM.

6. Cybersecurity concepts and working definitions

Asset and lifecycle foundations

Asset hygiene — The condition of an asset record and the asset itself: known owner, accurate identity, authorized status, supported software, current patches, secure configuration, appropriate access, valid warranty or maintenance, and

proper retirement. Good hygiene reduces uncertainty and prevents small data-quality problems from becoming security gaps.

Inventory — A maintained list of assets and their attributes. A useful inventory is authoritative enough to support decisions, not just a collection of names and serial numbers. It should include ownership, location or environment, criticality, relationships, software and versions, exposure, lifecycle state, and last-seen evidence.

Device types — The categories an organization uses to classify assets. Common types include laptops, desktops, servers, virtual machines, containers, mobile devices, routers, switches, firewalls, wireless access points, printers, storage, IoT, OT, medical devices, cloud resources, SaaS applications, and identities. Classification makes policy and prioritization possible.

Crown jewel — An asset, system, service, process, or data set whose compromise would cause disproportionate harm. Examples include payment systems, identity providers, production databases, patient-data systems, source-code repositories, domain controllers, and systems that control physical operations. Crown jewels should receive stronger ownership, monitoring, segmentation, backup, and recovery expectations.

Obsolescence — The condition of being outdated, difficult to support, incompatible with current standards, or no longer appropriate for the organization's needs. Obsolescence is not identical to vulnerability, but it tends to increase exposure because old products may lack fixes, modern encryption, vendor support, or skilled maintainers.

Maintenance — The planned work that keeps an asset reliable and secure: updates, repairs, configuration review, capacity management, certificate renewal, backup testing, support contracts, warranty coverage, and preventive replacement.

End of life (EOL) — A vendor or product lifecycle milestone after which the product is being retired or is no longer sold as part of the normal portfolio. EOL does not always mean support stops on the same day.

End of support (EOS) — The point after which the vendor no longer provides normal technical support, security fixes, or assistance. Some vendors use "end of service" for a similar idea.

End of maintenance (EOM) — The point after which regular maintenance releases, patches, or corrective updates stop. Vendors use these terms inconsistently, so contracts and product notices should be checked rather than assuming EOL, EOS, and EOM are interchangeable.

Vulnerability and exposure management

Vulnerability — A weakness in an information system, procedure, internal control, configuration, or implementation that could be exploited or triggered by a threat source. A vulnerability is not automatically a breach; risk depends on exploitability, exposure, asset criticality, compensating controls, and potential impact.

NVD — NIST's National Vulnerability Database, a U.S. government repository of standards-based vulnerability-management data. It enriches CVE information with product identifiers, references, impact metrics, and other data. NVD is important input, but it should be used alongside vendor advisories, CISA's KEV catalog, exploit intelligence, asset context, and local validation.

Patch management — The repeatable process of discovering missing patches, assessing urgency, testing or staging changes, deploying them, handling exceptions, and verifying that the vulnerability was actually reduced. A mature program prioritizes by business impact and exploitation, not by CVSS score alone.

Data breach — An incident in which unauthorized parties access, disclose, alter, destroy, or exfiltrate protected or sensitive information. Asset visibility cannot prevent every breach, but it reduces uncertainty about what exists, where sensitive data lives, which systems are exposed, and who must act during containment and recovery.

Perimeter — The boundary around a network, system, or environment where an attacker may attempt to enter, cause an effect, or extract data. Traditional security relied heavily on a fixed perimeter. Cloud, remote work, SaaS, mobile devices, partners, and BYOD make that boundary porous, which is why modern security focuses more on identities, devices, applications, and data.

Auto-remediation — An automated action that corrects a known condition, such as opening a ticket, isolating a device, changing a configuration, removing unauthorized software, applying a patch, disabling an exposed account, or requesting firewall changes. It should be governed by confidence, approval

thresholds, rollback, testing, and evidence of completion. “Automated” is not the same as “safe by default.”

Due diligence — The structured investigation performed before a decision such as an acquisition, supplier onboarding, major technology purchase, insurance renewal, or compliance attestation. IT and cyber due diligence asks what assets exist, how they are protected, which vulnerabilities and obligations remain, and what liabilities would transfer.

Operating model and service terms

MSP — A managed service provider that operates technology or security services for customers. MSPs need multi-customer visibility, reusable benchmarks, delegated ownership, standardized workflows, and strong tenant separation. A platform that lets an MSP compare posture against customer-specific rules can turn asset data into a repeatable service.

True SaaS — Software delivered and operated by the provider as a cloud service, usually with subscription access, provider-managed upgrades, elastic infrastructure, centralized security operations, and no customer-managed application server. “Agentless” and “SaaS” are different properties: a SaaS platform can still require endpoint agents, and an agentless platform can still use connectors or collectors.

Inbound integration — Data flowing into a platform from another system. Examples include Active Directory, cloud providers, endpoint tools, vulnerability scanners, CMDBs, ticketing systems, firewalls, and network discovery tools.

Outbound integration — Data or actions flowing from a platform to another system. Examples include tickets, patch jobs, firewall policy changes, reports, CSV exports, notifications, and updates to a CMDB. The value of an integration is highest when it closes the loop from finding to verified action.

Network protocols — The agreed rules that allow devices and services to communicate. In agentless discovery and asset assurance, relevant examples include TCP/IP and UDP for transport, ICMP and ARP for reachability and local discovery, DNS for naming, DHCP for address assignment, HTTP/HTTPS for web and API access, SSH for Unix administration, RDP and WinRM/WMI for Windows administration, SNMP for network devices, LDAP and Active Directory protocols for identity and directory context, and SMB for Windows file and management

services. Protocol visibility helps identify devices, but protocol exposure should never be mistaken for authorization.

ROI (return on investment) — The value received compared with the cost of the program. For ITAM and cybersecurity, ROI can include avoided breach loss, reduced audit effort, reduced tool overlap, lower license waste, fewer emergency fixes, faster incident response, better use of staff time, and more accurate procurement. It is best measured with a baseline rather than with a generic savings claim.

KPI (key performance indicator) — A metric that shows whether the program is achieving its intended outcome. Useful asset-assurance KPIs include inventory coverage, percentage of assets with an owner, duplicate rate, stale-record rate, unknown-asset count, unsupported-asset count, KEV exposure, patch SLA performance, mean time to remediate, percentage of crown jewels with current evidence, audit-preparation hours, and verified remediation rate.

7. Compliance and security standards

These terms are related but not interchangeable.

- **CISA** is the U.S. Cybersecurity and Infrastructure Security Agency. CISA publishes guidance and maintains the Known Exploited Vulnerabilities catalog, which identifies vulnerabilities exploited in the wild and is intended as an input to prioritization. CISA also publishes Cybersecurity Performance Goals that emphasize maintained asset inventories and accountability.
- **CISO** means Chief Information Security Officer. It is a role, not a standard or tool. The CISO is typically accountable for security strategy, risk decisions, governance, investment, policy, incident readiness, and communicating security posture to leadership.
- **HIPAA** — not “HIPPA” — is a U.S. law with privacy and security requirements for protected health information. The HIPAA Security Rule addresses safeguards for electronic protected health information. An asset inventory supports HIPAA evidence and risk management, but having an inventory alone does not create HIPAA compliance.
- **ISO/IEC 27001** is an international standard for an information security management system. Certification can demonstrate that an organization

operates a risk-management system meeting the standard's requirements. It is not a product certification and it does not guarantee that the organization cannot be breached.

- **NIST CSF 2.0** is a voluntary framework for managing cybersecurity risk. Its asset-management outcomes include maintaining inventories of hardware, software, services, systems, suppliers, data, and lifecycle state, and identifying and recording vulnerabilities.
- **SOC 2** is an attestation framework focused on controls relevant to trust-service criteria. It is separate from ISO/IEC 27001 and HIPAA, although organizations often map overlapping evidence and controls across them.

The practical lesson is that compliance is an **evidence and control-management problem**. A company needs to know what should be protected, which controls apply, who owns them, whether they operate, and how to prove that they operated during the relevant period.

8. SOAR, CAASM, and the role of ITAM in Zero Trust

SOAR — Security Orchestration, Automation, and Response — coordinates security tools and automates response playbooks. A typical playbook may ingest an alert, enrich it with asset and identity context, assess severity, create a ticket, isolate a device, request approval, and record the outcome. ITAM makes SOAR more reliable because automation needs accurate asset identity, owner, criticality, location, and lifecycle data.

CAASM — Cyber Asset Attack Surface Management — is the practice of bringing together data from security and IT tools to discover, normalize, correlate, and continuously analyze cyber assets and exposures. CAASM helps answer questions such as:

- Which assets are missing from endpoint protection?
- Which internet-facing systems contain known exploited vulnerabilities?
- Which cloud resources are owned by a departed employee?
- Which crown-jewel systems lack current patches or an accountable owner?
- Where do CMDB, directory, vulnerability, and cloud inventories disagree?

ApexaiQ overlaps with CAASM because it emphasizes unified asset visibility, enrichment, prioritization, and action. The product's positioning also adds lifecycle hygiene, obsolescence, compliance, and the ApexaiQ score as a management layer.

ITAM and Zero Trust

NIST describes Zero Trust Architecture as moving defenses away from static network perimeters and toward users, assets, and resources. Zero Trust assumes that network location or ownership does not create implicit trust; authentication and authorization are evaluated before access is granted.

ITAM is not Zero Trust by itself, but it supplies essential context for it:

1. **Identify the device.** A policy cannot make a good access decision about an unknown device.
2. **Assess posture.** Version, patch, encryption, EDR coverage, configuration, and lifecycle status influence whether access should be allowed.
3. **Understand criticality.** Crown-jewel systems require tighter access, segmentation, and monitoring.
4. **Map ownership and relationships.** Security teams need to know who can approve access and which business service would be affected.
5. **Continuously reevaluate.** A device that was healthy yesterday may become vulnerable, unsupported, or compromised today.
6. **Apply least privilege.** Accurate identity and asset data helps restrict users, workloads, and services to the minimum required access.
7. **Prove control.** Inventory, policy, and remediation evidence supports audits and incident investigations.

The simple relationship is: **Zero Trust decides whether access should be trusted; ITAM helps establish what the asset is, who owns it, how risky it is, and what policy should apply.**

9. A practical operating workflow

A usable program can be run as a cycle rather than a one-time inventory project:

1. **Discover** assets through integrations, APIs, network protocols, cloud accounts, directories, scanners, and existing management tools.
2. **Normalize** names, identifiers, owners, locations, device types, and relationships; remove duplicates.
3. **Enrich** records with vulnerabilities, KEV status, software versions, warranties, EOL/EOS/EOM dates, business criticality, and compliance requirements.
4. **Prioritize** by business impact, exposure, exploitability, asset criticality, and the presence of compensating controls.
5. **Act** through a ticket, change request, patch job, configuration change, isolation action, replacement, access review, or retirement.
6. **Verify** that the action happened and that the exposure or hygiene issue actually changed.
7. **Report** progress with score movement, KPI trends, exceptions, evidence, and remaining risk.

A strong first implementation does not try to fix every issue at once. It starts with the assets that can create the greatest operational or security harm: internet-facing systems, identity infrastructure, sensitive-data stores, unsupported technology, known-exploited vulnerabilities, and crown-jewel services.

10. Questions to ask before evaluating ApexaiQ or any alternative

- Which asset classes are discovered natively, and which require an integration?
- How are duplicates, stale records, shared devices, and changing IP addresses reconciled?
- How does the platform identify owners and business criticality?
- Which protocols and credentials are required for agentless discovery?
- How are cloud, SaaS, OT, IoT, mobile, and third-party assets handled?
- Which vulnerability sources are used, and how are NVD, vendor advisories, CVSS, EPSS, and CISA KEV considered?
- How are EOL, EOS, EOM, warranty, and maintenance dates maintained?

- Can the customer define benchmarks and exceptions?
- What does the score measure, and can its drivers be explained to an auditor?
- Does remediation include verification, or only ticket creation?
- How are tenant boundaries, data retention, encryption, and access controls implemented?
- Which data leaves the environment, and how are integrations authenticated?
- What evidence supports ROI claims and customer case-study outcomes?

Sources and further reading

ApexaiQ

- [ApexaiQ homepage](#)
- [ApexaiQ score and platform overview](#)
- [ApexaiQ banking case study](#)
- [ApexaiQ MSP overview](#)

Standards and public guidance

- [NIST SP 1800-5: IT Asset Management](#)
- [NIST Cybersecurity Framework 2.0](#)
- [NIST SP 800-207: Zero Trust Architecture](#)
- [NIST National Vulnerability Database](#)
- [CISA Known Exploited Vulnerabilities Catalog](#)
- [CISA Cybersecurity Performance Goals](#)
- [HHS Summary of the HIPAA Security Rule](#)
- [ISO/IEC 27001:2022](#)

Competitor case studies

- [Axonius: global specialty retailer](#)

- Armis: medical technology and manufacturing
- runZero: Capgemini
- JupiterOne: American financial-services company.